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Mukunuwenna

Alternanthera sessilis

Mukunuwenna is the commercially grown and most popular leafy vegetable type in the market. This is popular due to its nutritional, medicinal and characteristic flavor. The crop is rich in vitamins, calcium and iron



Available Types

1. Vine type
2. Bushy type

Vine type : this type is grown commercially

- Type with oval shaped leaves and light green colored stem. Most popular type in the market (Piliyandala type)
- Type with oval shaped leaves, brown and green colours mixed stem. Internodes are very short, mostly grown in pannipitiya, Arawwala areas
- Type with long oval shaped leaves and brown coloured stem.
- Type with large, long leaves, internodes are long with brown or green stem
- Type with lanceolate leaves and green or brown coloured stem (Weda Mukunuwenna).
- Type with long, narrow leaves, internodes are long

Climatic requirements/ Areas suitable for cultivation

Low country wet zone is most suitable for the cultivation. If irrigation facilities are available it can be grown in any other areas in the country

Soil

Well drained low lands are more suitable for the cultivation. If the irrigation facilities are available, the crop can be grown in uplands. Suitable pH range is 6-7.

Propagation

Stem cuttings are used for the propagation. 10-12 cm or longer cuttings can be planted or broadcasted.

Land preparation

Raised beds for low lands (paddy fields) and sunken beds for highlands are more suitable.

Soil is ploughed and allowed weeds to decompose. Then the fields are filled with water and take a fine filth and levelled. Drains are prepared to facilitate the drainage.



Planting

Stem cuttings are planted in specified spacing or randomly. Single cutting is planted at one point.

Spacing

Stem cuttings can be planted randomly or with the recommended spacing. Inter row spacing is 20 cm and intra row spacing is 10 cm

Fertilizer

Organic fertilizer and chemical fertilizers are applied poultry manure is the commonly used organic fertilize in commercial cultivations. Nematode problem can be minimized by using poultry manure.

Fertilizers for 1000m² (1/4 AC) area

-organic fertilizers 01 ton

-chemical fertilizers

Time of Apply	Urea kg	TSP kg	MOP kg
Basal – Applied 1-2 days before planting	9.0	13.5	10.0
Top dressing – After first harvesting	9.0	–	–
Top dressing – 2 nd harvesting and followed it at each harvesting	5.5	–	1.5
Top dressing – at every 6 months	–	6.5	–

Water supply

Irrigation is done frequently to maintain the moisture at the top layer of the bed. Water lodging should be avoided.



Weed Control

During the land preparation weeds should be controlled as much as possible. Hand weeding should be practiced throughout the cropping cycle

Disease Management

▲ 1. Brown leaf spot

Causal agent : This is caused by *Cercospora* sp.

Disease Transmission : This fungus is spread by soil and wind with infected plant parts.

Disease symptoms : Red or purplish brown spots appear on the leaves and spread throughout the leaf. Later the leaf turns to yellow color and dies.

Control of Leaf Spot Diseases

- If the disease is spreading intensively, harvest quickly and remove the diseased plant parts and burn them.
- Provide proper nutrition for cultivation.
- Avoid unnecessary application of nitrogenous fertilizers
- Following crop rotation
- Use healthy planting material for replanting.

In home gardening and organic farming

Liquid inoculation can be used to control *Trichoderma asperellum* leaf spot disease in home gardens and organic farming.

- Treatment of planting material and planting
 - *Trichoderma asperellum* (*Trichoderma asperellum*) liquid infusion (100 ml in 10 liters of water) immersed for 10 minutes and planted.
- Spray the *Trichoderma asperellum* solution every week after planting to thoroughly wet the plants and soil (pre-harvest period 7 days).

▲ 2. White rust disease

This is caused by *Albugo* sp.

Disease Transmission : This disease spread by soil and wind with infected plant parts. ^

Disease symptoms

- White spots appear on the underside of the leaves. White powdery fungal spores are produced in spots on the underside of leaves. The base of the plant is larger than normal. The leaves of the apical bud shrivel to the underside and stop growing. Damaged leaves turn to the yellow color and then fall off.

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Harvesting

First harvest can be obtained at four weeks after planting. Harvesting can be done at every 3-4 weeks after first harvesting.

Yield

1000-1200kg of harvest can be obtained at once from 1000 m² field.

By maintaining the field properly the crop can be harvested for two years. (8-10 picks in one year)\

At about 2 years the yield is reduced and the crop should be removed. Crop rotation should be practiced without planting the same crop again in the field.



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